THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — SCIENCE · HARD LEVEL UNIT TEST SET 1

Chapters 5–8: **Measurement of Length and Motion, Materials Around Us, Temperature and its Measurement, A Journey through States of Water** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. A rope is 3.2 m long. After cutting off pieces of 45 cm, 55 cm and 60 cm, the remaining length is: 1

(a) 100 cm **(b)** 200 cm **(c)** 160 cm **(d)** 320 cm

2. A student heats the ends of four rods P, Q, R, S, each with a wax-fixed pin at the same distance. The pins fall at 10 s, 25 s, 40 s and 60 s. The best conductor is: 1

(a) S **(b)** Q **(c)** P **(d)** R

3. Thermometer A reads 0 °C in melting ice and 100 °C in boiling water. Thermometer B reads 2 °C and 98 °C. For the same hot water at actual 50 °C, B would most likely read: 1

(a) 54 °C **(b)** 48 °C **(c)** 52 °C **(d)** 50 °C

4. When ice at 0 °C is heated, the temperature stays at 0 °C till all ice melts. This shows that: 1

(a) ice cannot be heated **(b)** heat is used to change the state
(c) the heater is weak **(d)** the thermometer is faulty

5. In a pendulum experiment, 20 oscillations took 36 s. A second student gets 18 s for 10 oscillations. Both agree because: 1

(a) time period is 1.8 s in both **(b)** time period differs
(c) he is wrong **(d)** 36 s vs 18 s differ

6. A boy tests four materials A, B, C, D with a torch. Light passes fully through A, partly through B, not at all through C and D. Which can be used for a window to see out and still keep privacy? 1

(a) C **(b)** A **(c)** B **(d)** D

7. Rod A conducts heat twice as fast as rod B. If wax pins fall from A in 20 s from the heated end, the same pin on B would fall in about: 1

(a) 40 s **(b)** 10 s **(c)** 5 s **(d)** 20 s

8. Water of 100 g at 0 °C is converted to ice at 0 °C. The mass of ice formed is: 1

(a) 100 g **(b)** more than 100 g **(c)** 0 **(d)** less than 100 g

9. A student measuring a pencil places the 2 cm mark of the scale at one end and reads 12.5 cm at the other end. Length is: 1

(a) 10.5 cm **(b)** 2.0 cm **(c)** 14.5 cm **(d)** 12.5 cm

10. Why does a heavy ship made of steel float? 1

(a) its shape displaces much water **(b)** steel is lighter than water
(c) ships have no weight **(d)** water is sticky

11. The best conductor-insulator pair for a cooking pot and its handle is: 1

(a) glass and iron **(b)** wood and plastic
(c) plastic and aluminium **(d)** aluminium and plastic

12. A student claims that the water in a cup disappeared. The correct statement is: 1

(a) it vanished **(b)** it became oxygen
(c) it was destroyed **(d)** it changed to vapour and mixed with air

13. **Assertion (A):** The SI unit of length is the metre. 1
Reason (R): 1 metre is equal to 100 centimetres.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

14. **Assertion (A):** Clear glass is a transparent material. 1
Reason (R): Light passes through transparent materials almost completely.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

15. **Assertion (A):** The sense of touch is not a reliable way to judge temperature. 1
Reason (R): Our skin can be misled by what we touched before, so the same water may feel hot or cold.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. A road runs 3.6 km. A boy walks at a step of 60 cm. How many steps does he take? 2

17. Explain why an empty plastic bottle floats but a bottle filled with sand sinks. 2

18. Two beakers contain 100 mL water at 70 °C and 700 mL water at 70 °C. Compare their temperatures and say if the bigger one has more heat. 2

19. Explain why a wet cloth kept in a closed polythene bag does not dry. 2

20. Convert 0.003 km into cm. 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. The circumference of a wheel is 2.2 m. How many turns will it make to cover 1.1 km? 3

22. Substance X is a shiny solid that does not dissolve in water, sinks in water and becomes hot at the other end when one end is heated. Name its class of material and give two reasons. 3

23. A student measures the temperature of boiling water as 98 °C in a hill station and 100 °C in a plain. Suggest a reason. 3

24. A tank of 2000 L loses 4 L per day by evaporation. If it is filled once a month (30 days) how much is lost, and what percentage? 3

OR

Describe an experiment to measure the length of a curved line of 1 metre length marked on a floor by two students. 3

25. Explain the importance of non-conducting handles in electrical tools. 3

OR

Describe why the lid of a tiffin box should be airtight to keep food warm. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. **Case study.** Sunita measured the length of her classroom using her footsteps and got 12 steps. Her friend Priya measured the same room and got 15 steps. They were confused, so their teacher in Bhopal asked them to use a measuring tape. The tape gave 9 m for both girls. 5

(i) Name the non-standard unit used by the girls.
(ii) Why did they get different values?
(iii) What is the SI unit of length?
(iv) Express 9 m in centimetres and explain why a measuring tape is better.

27. Describe an experiment to find which of three liquids (water, oil, milk) cools fastest. Explain controls, measurements and how to present results. 5

OR

Design an experiment to find which of the three places (sunny, shady, windy) is best for drying clothes and how to compare. 5

END OF QUESTION PAPER — Total: 50 Marks

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